

Submission Process for MiniProject 2 Phase3

Only submit three things:

1. Report (PDF, 4–6 pages)
2. GitHub Repository Link → This is the same as what we used for Phase 2 with your updates
3. Slides that you presented to the class (with updates if any)

Do NOT upload code to Canvas

Canvas must contain:

- Report (PDF)
- SLIDES (.pptx or .pdf)
- GitHub link (accessible and public)

GitHub Repository

Your repository must include:

- Agent implementation
- Execution scripts (single entry point required just one script runs everything)
- EDA interaction scripts (TCL / Python / shell)
- Sample inputs (or instructions)
- Example outputs (logs, reports, metrics)
- README (required)

README

Your README must allow the instructor/grader to reproduce your results.

It must include:

- Setup instructions (dependencies, environment)
- Exact commands to run the system
- Input/output description
- Expected results (for verification)
- Brief workflow description
- How to run the hidden testcases

The grader should be able to: clone → run → reproduce results

Execution Requirement

- Provide a single command/script to run your pipeline
- Workflow must be automated and reproducible. No manual steps

Report PDF (4–6 pages)

Formatting:

4–6 pages max, single column, single line spacing, 10 pt font, 1-inch margins. Include the following 10 section headings exactly:

- **Motivation for the Problem:** Briefly motivate the problem and its importance in EDA. Focus on why **automation alone is insufficient** and why **AI/agent-based approaches are needed**.
- **Problem Definition:** Clearly define the ICLAD (or modified version or different) task you are addressing, including:
 - Inputs
 - Outputs
 - Objectives
- **System Overview:** Provide a clear end-to-end view of your system:
 - Input → Agent → EDA Tool → Feedback → Output
 - Include a simple diagram if possible. This section should help the reader understand your full pipeline before diving into details
- **Agent Architecture:** (components, design, differences from prompting)
 - Describe your agent design, including key components:
 - Planner / decision module
 - Executor (tool invocation)
 - Parser (metrics extraction)
 - Feedback loop / iteration
 - Explain:
 - How decisions are made
 - How the system operates end-to-end
 - Clearly distinguish your approach from **one-shot prompting**:
 - Iterative reasoning
 - Tool-in-the-loop execution
 - Adaptive decision-making
- **Tools & Frameworks (EDA tools, LLMs, agent frameworks)**

List all tools used, including:

 - EDA tools (e.g., OpenROAD)
 - LLMs (e.g., GPT, Gemini)
 - Agent frameworks or custom infrastructure

Clearly state the **role of each tool** in your system

- **Context Engineering**

Describe:

- What information is provided to the agent (e.g., timing reports, logs, DEF)
- How context is structured:
 - raw text / JSON / summaries
- How you manage:
 - context size
 - relevance
 - noise filtering
- **Improvements from Phase 2:** Describe the difference or your progress from Phase 2
- **Results**
 - Present results on **multiple problems**
 - **For google Problems identify the exact correct RTL from all the 31 RTL for all visible testcases**
 - Include:
 - Metrics (WNS, TNS, runtime, violations, etc. for spec2GDS for example)
 - Provide **analysis**, not just numbers:
 - Why did it improve?
 - Where did it fail?
- **Challenges, Limitations, Improvements**
 - Discuss:
 - Key challenges encountered (technical/system-level)
 - Limitations of your current approach
 - Concrete improvements for future work
 - Focus on:
 - scalability
 - robustness
 - generalization
 - Runtime
- **AI Usage & Insights:** Clearly describe the role of AI in your system:
 - decision-making
 - script generation
 - debugging
 - optimization guidance

Provide **deep insights**, not just descriptions:

- Where AI works well
- Where it fails (hallucinations, incorrect commands, poor reasoning)
- Where human intervention was required
- **Experience with AI:**
Reflect on your experience using AI in this project.

Discuss:

- How AI was used during development (coding, debugging, understanding tools)
- Impact on productivity

- Limitations encountered
- What you learned about using AI as an engineer